

Algorithm: Image Enhancement Using Histogram Equalization

Algorithm

Step 1: Start the program.

Step 2: Read the input image using OpenCV.

Step 3: Convert the input image into a grayscale image using `cv2.cvtColor()`.

Step 4: Calculate the histogram of the grayscale image by counting the frequency of each pixel intensity from 0 to 255.

Step 5: Apply histogram equalization to redistribute the pixel intensity values using `cv2.equalizeHist()`.

Step 6: Display the original, grayscale, and histogram-equalized images using Matplotlib.

Step 7: Plot the histogram of the grayscale image.

Step 8: Plot the histogram of the equalized image and compare the intensity distributions.

Step 9: Stop the program.

Pseudocode

START

Read input image

Convert input image to grayscale

Apply histogram equalization

 Equalized Image = Equalize Histogram(Grayscale Image)

Display:

 Original Image

 Grayscale Image

 Equalized Image

Calculate and display:

 Grayscale Histogram

 Equalized Histogram

STOP

Result

Thus, the input image was converted into a grayscale image and its contrast was enhanced successfully using histogram equalization.